

CUSTOMER CHURN ANALYSIS & Prediction

DEPI Graduation Project Documentation

Full Technical Documentation — All Roles & Deliverables

Python 3.14.1 | LightGBM | Microsoft SQL Server | Streamlit

End-to-End Machine Learning Pipeline for Telecom Customer Retention

1. Project Overview

This project builds a complete end-to-end Machine Learning pipeline to predict customer churn for a telecom company. The goal is to identify customers likely to leave before they do, enabling proactive retention strategies.

Property	Value
Dataset	customer_churn.csv
Total Customers	999,999
Features	32 columns
Target Variable	churn (0 = Stayed, 1 = Left)
Churn Rate	9.92%
Best Model	LightGBM
F1-Score (Churn)	0.257
Language	Python 3.14.1
Database	Microsoft SQL Server 16.0
Dashboard	Streamlit

2. Project Structure

Path	Description
data/raw/customer_churn.csv	Original raw dataset — never modified
data/processed/clean_data.csv	Cleaned, encoded, scaled dataset
notebooks/01_data_preprocessing.ipynb	Initial data exploration and Full preprocessing pipeline
notebooks/02_model_training.ipynb	Model training & comparison
notebooks/03_model_evaluation.ipynb	Metrics & confusion matrix
notebooks/04_sql_analysis.ipynb	SQL queries & integration
database/schema.sql	SQL Server table schema
database/load_data.py	Load CSV to SQL Server
database/queries/	5 analytical SQL queries
models/lgbm_model.pkl	Trained LightGBM model

Path	Description
models/scaler/scaler.pk1	Fitted StandardScaler
app/streamlit_app.py	Interactive dashboard
reports/	5 PDF reports
requirements.txt	Python dependencies
README.md	Project overview

3. Team Roles & Deliverables

Role	Responsibilities	Deliverables
Data Engineer	Cleaning, encoding, scaling, pipeline	clean_data.csv, preprocessing notebook
Data Scientist	Model training, evaluation, selection	lgbm_model.pkl, training notebooks
Data Analyst	EDA, visualizations, insights	EDA notebook, eda_report.pdf
SQL Engineer	Database design, queries, analysis	schema.sql, 5 queries, sql_report.pdf
BI Developer	Dashboard, prediction interface	streamlit_app.py

4. Data Engineering Pipeline

The preprocessing pipeline transforms the raw 32-column dataset into a clean, model-ready format through 6 sequential steps.

Step	Action	Result
1	Load raw data	999,999 rows × 32 columns
2	Drop customer_id, signup_date	30 columns remaining
3	Fill missing values with Median	5 columns fixed — 0 nulls
4	Ordinal Encoding	contract, education, paperless_billing
5	One-Hot Encoding	gender, marital_status, payment_method
6	StandardScaler	13 numerical columns scaled
7	Train/Test Split (80/20)	799,999 train 200,000 test

5. Machine Learning Models

Four classification models were trained and evaluated. The primary challenge was class imbalance (9:1 ratio), addressed using `class_weight='balanced'`.

Model	F1-Score	Recall	Precision	ROC-AUC	Status
LightGBM	0.257	0.636	0.161	0.635	✔Selected
Logistic Regression	0.256	0.642	0.160	0.635	Runner-up
XGBoost	0.253	0.603	0.160	0.627	Tested
Random Forest	0.001	0.000	0.545	0.500	Needs tuning

Why LightGBM was selected:

Best overall F1-Score and ROC-AUC. Significantly faster on 1M+ rows compared to other tree-based models. Production-ready with strong support for class imbalance.

6. SQL Database & Analysis

The dataset was loaded into Microsoft SQL Server for structured querying and business intelligence analysis.

Property	Value
Server	OMAR-ZIZO (SQL Server 16.0.1180)
Database	customer_churn
Table	dbo.customers
Row Count	999,999
Column Count	38 (after encoding)
Queries Executed	5 analytical queries

Key SQL Findings

Query	Finding
Overall Churn Rate	9.92% — Class imbalance of 9:1
Churn by Contract	Month-to-Month: 26.51% One Year: 12.65% Two Year: 5.65%

Query	Finding
High-Risk Customers	100 retained customers with high complaints & low satisfaction
Churn by Charges	Minimal variation — pricing not primary driver
Churn by Tenure	New customers (<1yr): 10.26% Loyal (3yr+): 9.33%

7. Key EDA Insights

#	Insight	Business Action
1	New customers (0-10 months) churn most	Launch onboarding & loyalty programs
2	Month-to-Month contracts = 26.5% churn	Offer discounts to switch to annual plans
3	High complaints = strong churn signal	Resolve complaints within 24 hours
4	Low satisfaction predicts churn	Regular satisfaction surveys + follow-up
5	Very High charges = 12% churn rate	Review pricing for high-charge segments

8. Streamlit Dashboard

An interactive web dashboard was built using Streamlit to visualize insights and provide real-time churn predictions.

Page	Description
☐ Overview	4 metric cards + key insights summary
☐ EDA	4 interactive charts — distribution, contract, complaints, satisfaction
☐ Prediction	Input sliders + LightGBM prediction + risk factors

How to Run

```
cd D:\project
streamlit run app/streamlit_app.py
```

9. Requirements & Installation

Python Version

Python 3.14.1

Install All Dependencies

```
pip install -r requirements.txt
```

Package	Version	Purpose
pandas	2.2.3	Data manipulation

numpy	2.1.3	Numerical computing
scikit-learn	1.5.2	ML pipeline & metrics
lightgbm	4.5.0	Primary ML model
xgboost	2.1.3	Alternative ML model
imbalanced-learn	0.12.4	SMOTE for class imbalance
matplotlib	3.9.3	Data visualization
seaborn	0.13.2	Statistical visualization
pyodbc	5.2.0	SQL Server connection
streamlit	1.40.2	Web dashboard
joblib	1.4.2	Model serialization

End of Project Documentation